

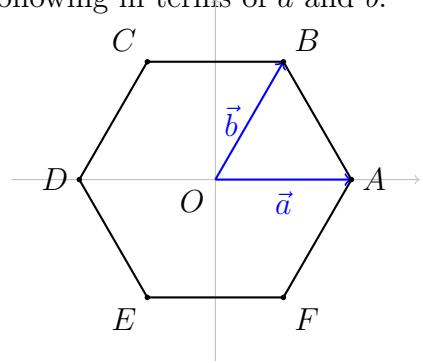
G11 PHYSICS

Assignment 01

Due: 14 March 2026

1. In the figure, A, B, C, D, E and F are vertices of a regular hexagon centered at the origin O . If $\vec{a} = \vec{OA}$ and $\vec{b} = \vec{OB}$, express each of the following in terms of $\vec{a}$ and $\vec{b}$:

- | | |
|----------------|--------------------------------------|
| (a) $\vec{AB}$ | (b) $\vec{BC}$ |
| (c) $\vec{AD}$ | (d) $\vec{CF}$ |
| (e) $\vec{AC}$ | (f) $\vec{BC} + \vec{DE} + \vec{FA}$ |



2. You are given three vectors:

$$\vec{a} = \hat{x} + 3\hat{y} - \hat{z}, \quad \vec{b} = 3\hat{x} - \hat{y} + \hat{z}, \quad \vec{c} = \hat{x} + 3\hat{y}$$

Find:

- | | | |
|-------------------------|-----------------------------|---|
| (a) $\vec{a} - \vec{b}$ | (c) $\vec{a} \cdot \hat{x}$ | (e) $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$ |
| (b) a_x | (d) $\vec{a} \cdot \vec{b}$ | |
3. Two vectors having equal magnitude A make an angle θ with each other. Find the magnitude of the resultant and its direction relative to the first vector.
 4. Two vectors of equal magnitude 2 units have an angle 60° between them. Find the magnitude of (a) the sum of the vectors and (b) the difference of the vectors.
 5. A force of 10 N acts on a particle along a direction making an angle 25° with the vertical. Find the component of the force in both the vertical and horizontal directions.
 6. Find the separation vector $\vec{\Delta}$ from the source point $(3, 6, 9)$ to the field point $(0, 6, 8)$. Determine its magnitude (Δ) and construct the unit vector $\hat{\Delta}$.
 7. A pilot flying at an air speed of 50 m s^{-1} wishes to travel due north. He knows that there is a 10 m s^{-1} wind blowing from west to east.
 - (a) In what direction should he head his plane?
 - (b) What will be the duration T of his flight if his destination is 160 km away?